Shared Tensor Guidance Super-resolution Network for Locally Overlapping Hyperspectral and Multispectral Remote Sensing Image Fusion

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Hyperspectral and multispectral image fusion aims to generate a high-resolution hyperspectral image (HR-HSI) from a low-resolution hyperspectral image (LR-HSI) and a high-resolution multispectral image (HR-MSI). However, existing methods often suffer from performance degradation in non-overlapping regions—i.e., areas covered only by HR-MSI—resulting in the loss of fine spatial details and spectral consistency. To address this issue, we propose a novel fusion method named STG-Net (Shared Tensor Guidance Super-resolution Network), based on core tensor guidance and enhanced super-resolution for non-overlapping regions. First, a deep Tucker decomposition network is employed to learn a shared core tensor and corresponding factor matrices from the overlapping regions. Second, a Multiscale Attentive Core Synthesizer (MACS), incorporating multi-scale residual blocks, dense channel attention, and spectral–spatial co-attention, is designed to generate a full-scene core representation under the guidance of the overlapping core tensor. Finally, an Adaptive Weighted Fusion Reconstructor (AWFR) is introduced to seamlessly integrate overlapping and non-overlapping regions. Experimental results on the Pavia University dataset under 4× downsampling demonstrate that the proposed STG-Net significantly outperforms existing approaches, effectively eliminating grid artifacts while preserving spectral fidelity.

Keywords—Hyperspectral and multispectral image fusion; Tucker decomposition; Core tensor guidance; Multiscale attentive core synthesizer; Adaptive weighted fusion reconstructor

I. INTRODUCTION

Hyperspectral images (HSIs) offer rich spectral information for remote sensing but are often limited in spatial resolution. A common solution is to fuse low-resolution HSIs (LR-HSIs) with high-resolution multispectral images (HR-MSIs) to produce high-resolution HSIs (HR-HSIs).

Existing fusion methods can be broadly categorized into two groups: model-driven (non-deep) approaches and deep learning-based approaches. Traditional non-deep methods, such as CNMF [1][2], are typically grounded in tensor decomposition. However, they often suffer from limited flexibility, high computational complexity, and difficulty in capturing complex nonlinear relationships. In contrast, deep learning-based methods demonstrate strong feature representation capabilities. Nevertheless, most supervised approaches rely heavily on large volumes of paired training data and exhibit limited generalization in real-world fusion scenarios. Moreover, deep learning models are often regarded as black boxes, lacking interpretability.

In recent years, the integration of deep learning with tensor decomposition has led to notable advancements, enabling more effective exploitation of spectral–spatial structural information. Representative methods include DTDNML [3], HyCoNet [4], and CuCANet [5], which generally assume full overlap between the LR-HSI and HR-MSI. However, these methods continue to encounter significant challenges when dealing with non-overlapping regions. Specifically, the absence of spectral constraints from the LR-HSI often results in the loss of fine spatial details and the introduction of spectral distortions.

Existing deep tensor decomposition–based methods effectively exploit spectral–spatial information in overlapping regions but exhibit limitations in non-overlapping areas: the core tensor is learned only from overlapping regions, while simple upsampling and hard stitching are applied elsewhere. This leads to two issues, (1)grid artifacts caused by directly concatenating high-quality overlapping reconstructions with coarse non-overlapping outputs; (2)and loss of spectral and spatial details due to the lack of LR-HSI constraints, resulting in distortion and boundary blurring. To address these issues, this paper proposes STG-Net (Shared Tensor Guidance Super-resolution Network), a method based on core tensor guidance and enhanced super-resolution for non-overlapping regions. The main contributions are as follows:

(1) Local-to-global core tensor modeling: a shared core tensor and corresponding factor matrices are first learned from overlapping regions with reliable spectral–spatial information, and then extended to the full scene via an enhanced synthesis network, enabling effective representation of non-overlapping regions lacking direct spectral constraints.

(2) Multiscale Attentive Core Synthesizer (MACS): a dedicated core generation module that integrates multi-scale residual learning, dense channel attention, and spectral–spatial co-attention. Guided by the overlapping-region core tensor, MACS generates a high-quality full-scene core tensor from HR-MSI, effectively recovering fine spatial structures while maintaining spectral fidelity.

(3) Adaptive fusion with consistency constraints: an Adaptive Weighted Fusion Reconstructor (AWFR) with a learnable spatial attention mechanism is proposed to seamlessly fuse overlapping and non-overlapping regions, alleviating discontinuities and eliminating grid artifacts. In addition, region-specific reconstruction losses and a style

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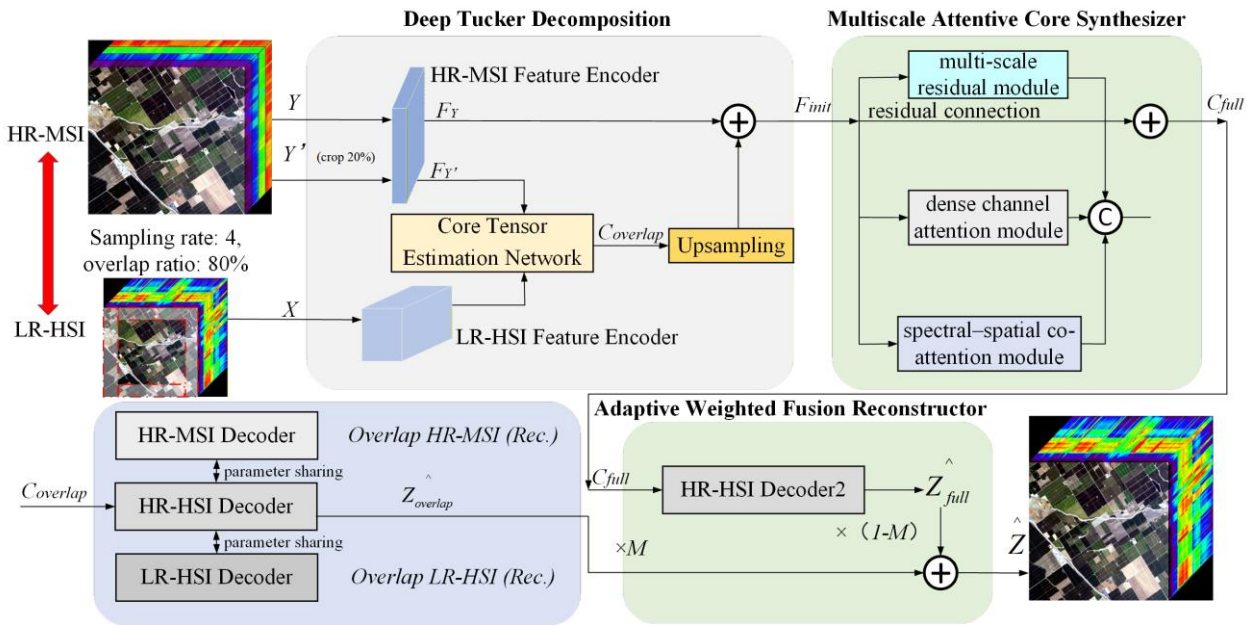

Fig. 1: Overall framework of the proposed method.

consistency constraint are introduced to enforce spectral consistency and suppress spectral drift across the entire scene.

II. METHODOLOGY

A. Overall Framework

The proposed method adopts a two-stage framework (Fig. 1). In Stage 1, an overlapping region (80%) is used to learn a shared core tensor and factor matrices via deep Tucker decomposition. In Stage 2, the full HR-MSI is leveraged to infer a complete core tensor for the non-overlapping region through a super-resolution module, while preserving spectral representations. Finally, an adaptive reconstruction strategy generates the HR-HSI, enabling a transition from local learning to global reconstruction.

B. Deep Tucker Decomposition in Overlapping Region

The HR-HSI Z is modeled using Tucker decomposition as

$$Z = C \times_1 W \times_2 H \times_3 S \tag{1}$$

where C is the core tensor and W, H, S denote spatial and spectral factor matrices.

The LR-HSI X and HR-MSI Y follow degradation models:

$$X = C \times_1 (P_1 W) \times_2 (P_2 H) \times_3 S + N_h \tag{2}$$

$$Y = C \times_1 W \times_2 H \times_3 (P_3 S) + N_m \tag{3}$$

where P_1, P_2 and P_3 are degradation operators.

In practice, LR-HSI and HR-MSI are first encoded into latent features, which are fused by a FeatureUNet with spectral attention to estimate the core tensor $C_{overlap}$. The factor matrices are implemented via lightweight convolutional mappings, enabling reconstruction of LR-HSI, HR-MSI, and overlapping HR-HSI.

C. Multiscale Attentive Core Synthesizer

To address the lack of spectral constraints in non-overlapping regions, a core tensor super-resolution network is introduced. The overlapping-region core tensor $C_{overlap}$ is first aligned with full-resolution HR-MSI features F_Y, and their fusion is used to predict a full-scene core tensor:

$$F_{init} = Up(C_{overlap}) \oplus F_Y \tag{4}$$

MACS refines F_{init} using three complementary components: a multi-scale residual module, a dense channel attention module, and a spectral–spatial co-attention module. Their outputs are concatenated and fused to produce the full-scene core tensor with a residual connection:

$$C_{full} = F_{init} \oplus F([F_{ms}(F_{init}), F_{dc}(F_{init}), F_{ss}(F_{init})]) \tag{5}$$

where $F_{\mathrm{ms}}(\cdot), F_{\mathrm{dc}}(\cdot)$ and $F_{\mathrm{ss}}(\cdot)$ denote the multi-scale feature extraction, dense channel attention, and spectral–spatial co-attention operations, respectively, and $F([\cdot])$ represents channel-wise concatenation.

D. Adaptive Weighted Fusion Reconstructor

Given the estimated full-scene core tensor C_{full}, the HR-HSI is reconstructed as

$$\hat{Z}_{full} = C_{full} \times_1 W \times_2 H \times_3 S \tag{6}$$

To mitigate blocking artifacts, an adaptive fusion strategy is adopted. The overlapping reconstruction $\hat{Z}_{overlap}$ is embedded into the full scene and combined using a learnable spatial weight map M:

$$\hat{Z} = M \times \hat{Z}_{overlap} + (1 - M) \times \hat{Z}_{full} \tag{7}$$

where $M \in [0,1]^{H \times W}$ is learned from core tensor and MSI features, ensuring smooth transitions between overlapping and non-overlapping regions.

E. Loss Function

The overall loss consists of reconstruction and consistency terms:

$$L = L_{rec} + \lambda_1 L_{degrad} + \lambda_2 L_{style} + \lambda_3 L_{manifold} \tag{8}$$

The reconstruction loss is defined as

$$L_{rec} = \left\| X - \hat{X} \right\|_1 + \left\| Y - \hat{Y} \right\|_1 + \left\| Z - \hat{Z} \right\|_1 \tag{9}$$

The degradation consistency loss enforces:

$$L_{\deg rad} = \left\| X - f_{PSF}(\hat{Z}) \right\|_1 + \left\| Y - f_{PSF}(\hat{Z}) \right\|_1 \tag{10}$$

A style consistency constraint is introduced as

$$L_{style} = \left\| Gram(\hat{Z}) - Gram(\hat{Z}_{overlap}) \right\|_1 \tag{11}$$

and manifold regularization [3] is applied to factor matrices to preserve spectral–spatial structure.

III. EXPERIMENTAL RESULTS

A. Data Preparation and Experimental Protocol

To validate the proposed method, experiments are conducted on the Pavia University (PaviaU) [6] dataset, acquired by the ROSIS sensor over Pavia, Italy. After removing corrupted bands, 103 spectral channels are retained.

Following the standard degradation model, a 128×128 HR-HSI patch is used as ground truth. The LR-HSI is generated via Gaussian blurring and 4× downsampling, while the HR-MSI is simulated using a spectral response function over four bands (blue, green, red, and near-infrared). Gaussian noise is added, with SNRs set to 25 dB (LR-HSI) and 40 dB (HR-MSI).

The network is trained using Adam [7] with a learning rate of 5×10^{-3} for 10,000 iterations, and linear decay starting from 3,000 iterations. Batch size is set to 1. Performance is evaluated using RMSE, PSNR, SAM, ERGAS, and SSIM, where ERGAS measures global distortion and SAM evaluates spectral consistency.

B. Comparison with State-of-the-Art Methods

Table I compares the proposed method with several representative approaches, including UDTN [8], MIAE [9], 3DT-Net [10] , and DTDNML [3]. All results are reported on the PaviaU dataset under the 4× degradation setting.

Our method achieves the best performance across all available metrics. Specifically, it attains a PSNR of 42.00 dB and an RMSE of 0.00802, which is less than half that of the baseline DTDNML (0.0158). The SAM value (2.25°) also significantly outperforms all competing methods, indicating superior spectral fidelity. While the ERGAS value (1.45) is slightly higher than that of DTDNML (1.24), this trade-off is

TABLE I QUANTITATIVE RESULTS ON PAVIAU DATASET (4× DOWNSAMPLING)

Method	RMSE↓	PSNR↑	SAM↓	ERGAS↓	SSIM↑
UDTN	0.0183	34.93	4.85	1.47	0.930
MIAE	0.0178	35.63	3.90	1.42	0.948
3DT-Net	-	41.13	2.48	1.90	-
DTDNML	0.0158	36.41	3.87	**1.24**	0.955
STG-Net	**0.0080**	**42.00**	**2.25**	1.45	**0.970**

acceptable given the substantial gains in spatial reconstruction and spectral angle accuracy. Compared with 3DT-Net, which reports a high PSNR (41.13 dB) but a relatively higher ERGAS (1.90), our method maintains a better balance between spatial and spectral quality.

C. Visual Analysis

Fig. 2 shows visual comparisons under the 4× setting, including reconstructed images and error maps. The ground truth exhibits clear structures and fine details. DTDNML produces relatively sharp results, but some structural details and boundary regions are less accurately reconstructed, as reflected by higher responses in the error maps.

In contrast, the proposed method generates cleaner boundaries and preserves finer textures, especially in complex regions. Its error map appears darker and smoother, indicating lower reconstruction errors. Overall, the proposed method achieves more accurate reconstruction with better spatial detail preservation and improved spectral fidelity.

D. Ablation Study

Ablation experiments are conducted on the PaviaU dataset under the 4× setting to evaluate each component. Four variants are compared: removing the Multiscale Attentive Core Synthesizer (w/o MACS), removing the Adaptive Weighted Fusion Reconstructor (w/o AWFR), removing the style consistency loss (w/o Style), and removing the manifold regularization (w/o Manifold). Table II reports the quantitative results.

The full model achieves the best performance across all metrics. Removing MACS causes the most significant drop (PSNR −3.8 dB, SAM +0.47°), confirming its essential role in recovering non-overlapping details. Disabling AWFR leads to visible grid artifacts and reduced PSNR (−2.1 dB). Excluding style loss degrades spectral fidelity (SAM +0.29°), while removing manifold regularization lowers structural

TABLE I QUANTITATIVE RESULTS ON PAVIAU DATASET (4× DOWNSAMPLING)

Variant	PSNR↑	SAM↓ (°)	SSIM↑
w/o MACS	38.20	2.72	0.953
w/o AWFR	39.90	2.41	0.958
w/o Style	41.20	2.54	0.966
w/o Manifold	41.55	2.31	0.962
Full model	42.00	2.25	0.970

consistency (SSIM −0.008). These results validate the contribution of each proposed module.

E. Metrics and Loss Analysis

Fig. 3 shows the convergence behavior under the 4× setting. In the upper panel, PSNR increases from 28.09 dB to 42.00 dB and SAM decreases from 10.05° to 2.25°, while RMSE drops to 0.00802 and SSIM improves to 0.970,

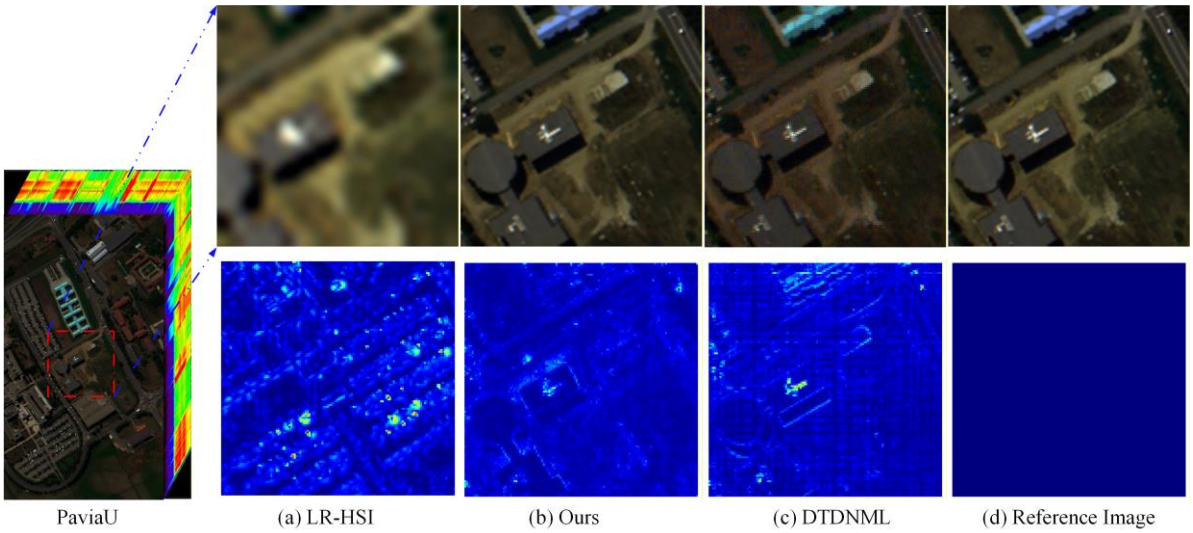

Fig. 2: Visual comparison under the 4× setting, including reconstructed images and corresponding error maps.

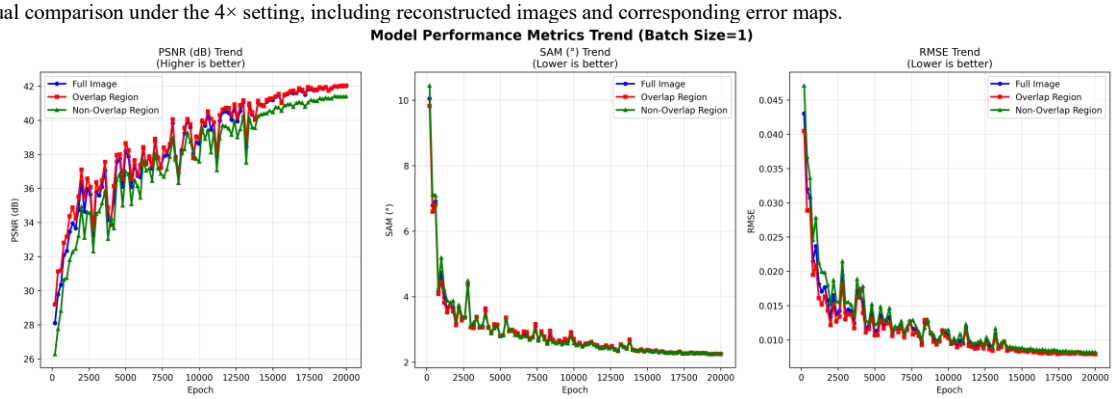

Fig. 3: Convergence analysis of different metrics under the 4× setting

indicating consistent enhancement in spatial and spectral reconstruction. Although the non-overlapping region initially lags behind, it gradually approaches the overlapping region, demonstrating the effectiveness of MACS in transferring spectral–spatial information.

IV. CONCLUSION

This paper addresses hyperspectral–multispectral image fusion in the presence of non-overlapping regions, where existing methods often suffer from spatial detail loss and spectral distortion. We propose STG-Net, a framework based on core tensor guidance and attentive core synthesis to achieve consistent high-resolution reconstruction across the full scene.

The method integrates deep Tucker decomposition with multiscale attentive synthesis and adaptive fusion, significantly improving spatial details and spectral fidelity while suppressing grid artifacts, particularly in non-overlapping areas. This demonstrates the effectiveness and robustness of combining tensor modeling with attention mechanisms for both interpretability and performance.

Future work will focus on real-world scenarios, efficiency optimization, and lightweight or self-supervised extensions, as well as improved cross-scene generalization.

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